Weekly Progress Report: Mediation Analysis & Package Validation

Xiaoni Xu

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Key updates include: - *Bug Fix*: Identified and resolved an ordering issue in the bootstrap loop where covariate-coefficient mismatching occurred due to duplicate IDs. - *New Feature: Exact Match Mode*: Implemented the `refit = TRUE` parameter in `cmest_pathcomprisk`, allowing for a full non-parametric bootstrap that matches the original script's logic. - *Model Validation*: Performed a head-to-head comparison of the original re-fitting method versus the package's default parametric approach.

Mode Comparison & Validation Results

To ensure the package behaves as expected, I compared two different bootstrap philosophies using a subset of the SHS Mimick data ($N = 500, B = 20$).

1. Parametric/Hybrid (Default)

This mode fits models once and then draws from the parameter distributions (`rmvnorm`) in each iteration. It is significantly faster and computationally stable.

2. Non-parametric (Exact Match)

This mode re-fits all models (mediators, dropout, and outcome) on a new resampled dataset in every iteration. This matches the original script's logic and captures uncertainty in the model-fitting process itself.

Comparison Results (N=500, B=20)

Effect	Old Method (Re-fitting)	New Package (Parametric)
Direct Effect (DE)	32.98	-9.24
Indirect Effect through Mediator (IEM)	13.79	-4.44
Indirect Effect through Dropout (IED)	-0.36	-0.16
Total Effect (TE)	55.71	8.08

Note: The results for IED are now aligned in direction and magnitude following the ordering bug fix.

Production Run Details

A large-scale production run is currently underway using the following parameters: - Sample Size (N): 7,500 - Bootstrap Samples (B): 1,000 - Mode: Exact Match (`refit = TRUE`) - Estimated Completion: Several hours (running in-process)

Model Definitions (Refit Mode)

The package now handles the re-fitting of these models automatically inside the bootstrap loop:

```r
# Example of the new 'refit' call in the package
results <- cmest_pathcomprisk(
    D = D_models,
    mreg = mreg_models,
    mvar = c("M1", "M2"),
    yreg = yreg_model,
    avar = "E",
    data = simData,
    nboot = 1000,
    refit = TRUE,                 # Enable Exact Match
    yreg_time = "time_to_event", # Time variable name
    yreg_event = "eventHappened" # Event indicator name
)
```

Next Steps

- Monitor and complete the $N = 7,500$ production run.
- Update the documentation for the `refit` parameter in the `cmcrhazard` package.
- Prepare a final summary report once the large-scale results are available.